Improving Satellite XCO_2 Measurements Accuracy: A Bayesian Bias Correction Approach Considering Spatiotemporal Bias Characteristics

Ruoxi Li, Xiang Zhou, Tianhai Cheng, Zui Tao, Ning Wang, Hongming Zhang, and Tingting Lv

***Abstract*— Measurements of column-averaged dry air mole fraction of CO_2 (XCO_2) from satellite contain systematic errors and regional scale biases are often induced by the limitations of the retrieval algorithm. Although the global linear bias correction (BC) model has successfully reduced some of these systematic errors, the unified correction formula and filter have proven too restrictive for certain regions, resulting in high remaining systematic errors. We propose a BC method through Bayesian optimal estimation considering spatiotemporal bias characteristics (ST-BCs) to address this issue. The prior distribution is provided by the global flux model, and the corrected samples are calculated by: 1) training a nonlinear model of satellite correction parameters and truth proxy based on XGBoost machine learning; 2) adjusting spatial characteristics bias based on spatial correlation of systematic errors; and 3) using the total carbon column observation network (TCCON) long-term measurement trend to correct temporal characteristics bias. The posterior correction value is iteratively calculated through Bayesian optimal estimation. We conducted experiments using OCO-2 V10 retrieval in the study areas in Europe, Asia, and North America, where TCCON coverage is extensive. The results demonstrated that the method effectively reduces regional bias and also significantly diminishes the correlation between state parameters and bias. The application of ST-BC enhances the correction accuracy of ACOS V10 products from 1.07 to 0.88 ppm. Particularly, the method reduces bias by 0.78 ppm for complex summer measurements. This study improves the accuracy of satellite retrieval XCO_2 and will advance our understanding of the global and regional carbon sources and sinks.**

***Index Terms*— Bayesian, bias correction (BC), mole fraction of CO_2(XCO_2), spatiotemporal characteristics.**

I. Introduction

CARBON dioxide (CO_2) is the main contributor to radiative IPCC [1], and hence, the impact of rising atmospheric CO_2 concentrations on future climate issues has attracted widespread attention [2], [3], [4], [5]. Following a long history of in situ measurements of CO_2 at key sites allowed us to understand the carbon cycle on continental scales [6], [7], [8]. In the era of space-based remote sensing, satellite measurements have improved our cognition of the global carbon cycle more dynamically and profoundly. The space-based measurements of atmospheric CO_2 with unique advantages have successfully quantified carbon sources and sinks globally [9], [10], [11], regionally [12], [13], [14], and even at the city scales [15], [16], [17]. Nevertheless, satellite retrieval column-averaged dry air mole fraction of CO_2 (XCO_2) inevitably contains bias induced by measurement noise, instrument calibration, spectroscopy, complex aerosols, and surface characteristics [18], [19], [20].

Bias correction (BC) is an important process to control systematic errors of ACOS retrieval products. Typically, a filter with an adjustable threshold is employed to confine the parameters within a range where they exhibit a linear relationship with the XCO_2 bias. Subsequently, multiple linear correction is applied to the state parameters that should not affect the retrieval, such as aerosols, surface characteristics, and surface pressure. This approach effectively reduces measurement uncertainty. With advancements in retrieval algorithms, various versions of BC products have been developed for GOSAT [19], [21], [22], [23], OCO-2 [24], [25], [26], [27], [28], and OCO-3 [27], [28]. However, at the regional scale, the globally unified correction formula and filter are too restrictive for certain regions [19], [29], [30]. Consequently, these correction products undoubtedly struggle to capture the regional-scale bias characteristics [31], failing to meet the accuracy requirements for regional bias.

We quantified the regional limitations of the linear correction method by comparing the total carbon column observation network (TCCON) and OCO-2 V10 2 Lite products. First, the root means square error (RMSE) between the corrected product and TCCON measurement is 2.03 $\pm$0.97 ppm, which means that the bias exceeds the application requirement of 1 ppm in certain regions, such as ET and PA in North America, PR and SO in Europe, and RJ in Asia [Fig. 1(a)]. Second, the correction parameters still exhibit much correlation with bias, resulting in residual systematic errors in certain regions. Notably, the average residual systematic error amounts to 1.63 ppm in North America [Fig. 1(b)].

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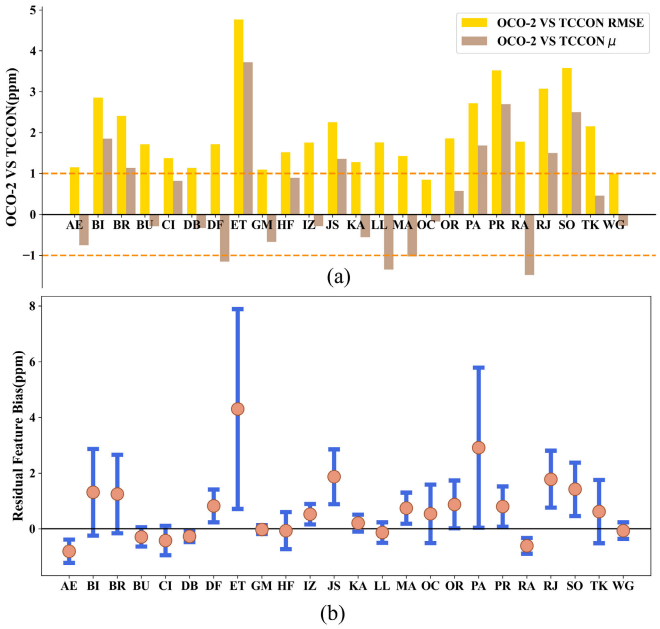


Fig. 1. (a) Shows the RMSE and μ of the OCO-2 V10 bias-corrected product from TCCON measurements. (b) Shows the residual systematic error of TCCON.

To address this issue, we proposed a Bayesian BC method considering spatiotemporal bias characteristics (ST-BCs). The corrected samples are calculated as follows:

1) Machine learning is used to construct a nonlinear functional relationship between regional correction parameters and truth proxy. Previous studies have used XGBoost machine learning to build a global correction model, improving the land measurement accuracy of V10 from 1.07 to 1.03 ppm [32]. The difference is that we have separately built XGBoost models for key regions in an attempt to achieve higher precision.
2) Satellite measurement system errors exhibit spatial heterogeneity and correlation under diverse measurement environments. By leveraging the spatial characteristics of regional bias, we enhance the accuracy of the correction process.
3) Long-time TCCON measurement series are used to construct a satellite time series reference benchmark to improve the correction accuracy on a time scale.

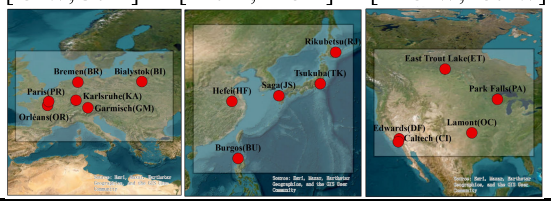
1) Follows the theoretical basis of systematic error correction and regionalizes it; 2) considers the spatial characteristics of regional deviations; and 3) utilizes the time series information of ground-based measurements. The correction values calculated by each idea are subject to uncertainty, and Bayesian optimal estimation can integrate multisource information.

II. Data and Preprocessing

A. Study Area

This study focused on North America, Asia, and Europe as research areas, where TCCON covers more worldwide, and determined the research scope by further analyzing the spatial correlation of regional XCO_2. Specifically, we used the spatial distribution of XCO_2 observed by OCO-2 in 2018 to construct a regional empirical semi-variogram using Gaussian iterative fitting (Table I). The nugget values of the three regions are all close to 0, indicating minimal spatial variability at very short distances. The base values provided insights into the strength of regional correlation, with Europe exhibiting the strongest correlation, followed by Asia and North America. The range characterized the spatial extent of correlation. By considering the range parameter alongside the location of TCCON measurements, we determined the precise scope of the study area.

TABLE I
This Is the Study Area

Region	Europe	Asia	N.America
Range	25.91	18.24	24.84
Sill	0.56	0.44	0.20
Extent	[40°N, 60°N] [-5°W, 30°E]	[15°N, 50°N] [110°E, 148°E]	[30°N, 60°N] [-125°W, -80°W]
Map			

B. ACOS V10 Product

ACOS V10 represents the fourth significant post-processing iteration of OCO-2 measurements, incorporating updates to the Level 2 full-physics retrieval algorithm [27]. The process from observed spectrum to XCO_2 product involves several steps, including measurement prescreening, algorithm retrieval, BC, and quality filtering. Initially, measurement prescreening removes those with XCO_2 errors that exceed theoretical predictions. Then, the ACOS algorithm is used to retrieve the radiation of the 0.76μm near-infrared O_2 A band, the 1.6 μm short-wave infrared weak CO_2 band, and the 2.05μm short-wave infrared strong CO_2 band into the XCO_2 standard product. Finally, the raw retrieval product undergoes systematic BC to produce Level 2 Lite product as in

$$\mathrm{XCO_{2Corrected}} = \frac{\mathrm{XCO_{2Raw}} - \mathrm{FOOT}[fp, \mathrm{mode}] - \mathrm{FEATS[mode]}}{\mathrm{TCCON_ADJUST[mode]}} \quad (1)$$

$\mathrm{XCO_{2Raw}}$ is raw retrieval of XCO_2; FOOT[fp, mode] denotes the bias associated with the differences in the footprint atmospheric path; TCCON_ADJUST[mode] links the satellite measurement to the TCCON scale; FEATS[mode] represents the most complex and crucial systematic error originating from the retrieval algorithm itself, and is eliminated using the multilinear formula described in

$$\mathrm{FEATS[mode]} = \sum_i c_i \left(p_i - p_{i,\mathrm{ref}}\right). \quad (2)$$

Here, p_i represents parameters that are theoretically independent of the retrieval bias; $p_{i,\mathrm{ref}}$ is the theoretical reference value; and c_i is the linear correction coefficient. Global unified multiple linear regression inevitably leads to limitations in regional BC, which is the key focus of this research.

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We used the XCO_2 retrieval products derived from OCO-2 ACOS V10 for the period spanning 2014–2018 and conducted footprint correction on the raw retrieval. The selection of correction parameters depends on the limitations of the retrieval algorithm. For the ACOS V10, this article focused on the following parameters for land measurements: 1) the discrepancy between the retrieval surface pressure after altitude adjustment and the prior surface pressure (dP_{frac}); 2) the integrated aerosol optical depth of coarse mode particles (DWS $=$ $\text{AOD}_{\text{dust}} + \text{AOD}_{\text{water cloud}} + \text{AOD}_{\text{sea salt}}$); 3) the difference between the retrieval CO_2 broad line and the prior "CO_2 grad del;" and 4) the fine aerosol optical depth AOD_{fine}. Additionally, we selected soundings whose quality flag is 0 of the land objects in our experiment to ensure the relevance and suitability of the selected data for our analysis.

C. Total Carbon Column Observation Network

The TCCON continuously measures high-resolution near-infrared spectra through ground-based Fourier transform spectrometers (FTSs). The retrieval of XCO_2 through the standardized GGG program has been widely validated by the World Meteorological Organization (WMO) and the errors are below 0.25% (∼1 ppm) [33]. We downloaded 16 TCCON datasets (Table II) and use OCO-2 soundings that we use OCO-2 soundings that are coincident with TCCON GGG2014 measurements in time (2 h) and space (2.5° Lat, 5° Lon) same as V10, performed a mean kernel correction on the TCCON observations, and corrected for the presence of atmospheric water vapor.

In this research, TCCON data are used as the only truth proxy to calculate the parameter BC coefficient for each single TCCON site and as the target value of XGBoost machine learning. Also, they serve as ground control points to construct a time series reference benchmark for satellite measurements.

D. Flux Models

Flux models are often used as truth proxies for BC and have larger spatial coverage compared to TCCON. CarbonTracker, developed by NOAA Global Monitoring Laboratory, is based on the global chemical transport model (TM5) and ensemble Kalman filter (EnKF) algorithm to assimilate ERA data. It provides a spatial resolution of 2° Lat × 3° Lon and a temporal resolution of 3 h [50], [51]. CAMS global greenhouse gas reanalysis data, released by the Copernicus Atmosphere Monitoring Service, is based on the integrated forecast system (IFS) radiative transfer model. It uses the atmospheric model 4-D variation (4D-Var) of physical and chemical laws to assimilate ECMWF data, offering higher spatial resolution (0.75°) and temporal resolution (3 h) [52], [53].

To assess the consistency of these flux models with TCCON measurements, we compared the residual error of CarbonTracker and CAMS from TCCON during 2015–2018, which were 0.14 ±1.11 ppm and 0.49 ±1.40 ppm, respectively (Fig. 2), both below 1.5 ppm. In our research, we used CarbonTracker and CAMS as prior datasets and filtered out the data beyond 3 σ in the time (3 h) and space (2.5°Lat, 5°Lon) range of OCO-2 soundings as the Bayesian prior distribution.

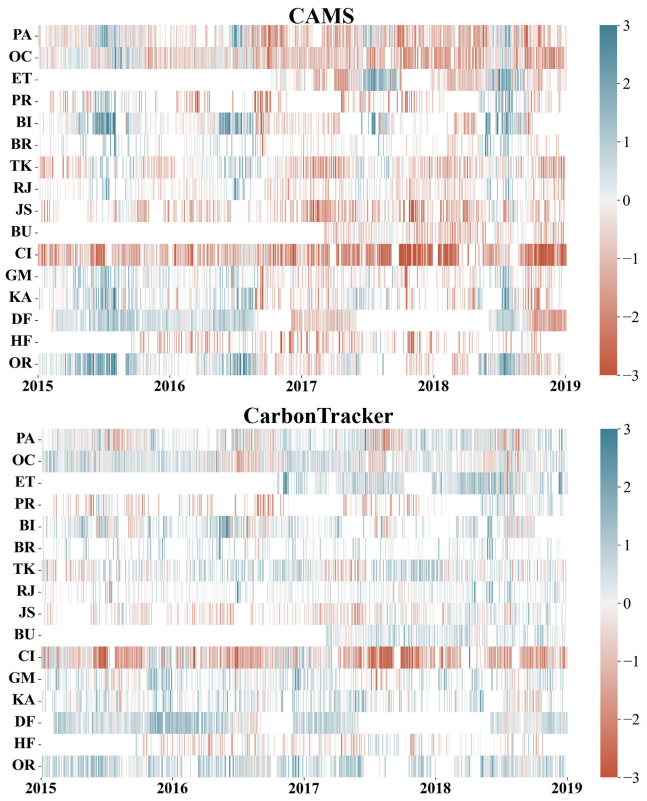


Fig. 2. Models versus TCCON.

III. ST-BC Methods

A. BC Based on XGBoost

We use the eXtreme gradient boosting (XGBoost) machine learning based on gradient boosting [54] to construct the relationship between the retrieval state parameters and the truth proxy.

XGBoost comprises a collection of iterative residual trees, each of which learns the residuals of the previous $N-1$ trees. The final predicted value $\hat{y}_i$ for a new sample is the sum of the predicted values $f_t(x_i)$ of each tree in (3). The objective function of XGBoost consists of a regularization term in (4) and a loss function in (5). λ and γ are hyperparameters of the regularization term. During tree fitting, the information gain is calculated by evaluating the sum of the gradient G of the loss function and the gain H of the left and right leaf nodes. We minimize the root mean square error between the target y_i and estimated value $\hat{y}_i$ as the loss function

$$\hat{y}_i = \sum_{t=1}^{n} f_t(x_i),\ f_t \in F \tag{3}$$

$$X_{\text{score}} = \frac{1}{2}\left[\frac{G_{\text{Left}}^2}{H_{\text{Left}}+\lambda} + \frac{G_{\text{Right}}^2}{H_{\text{Right}}+\lambda} - \frac{\left(G_{\text{Left}}+G_{\text{Right}}\right)^2}{H_{\text{Left}}+H_{\text{Right}}+\lambda}\right] - \gamma \tag{4}$$

$$\text{Loss} = \frac{1}{N}\sum_{i=1}^{N}\left(y_i - \hat{y}_i\right)^2. \tag{5}$$

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TABLE II
DETAILS OF 16 TCCON SITES

Site	Lat/°	Lon/°	Access Dates	REF
Park Falls(PA)	45.94	-90.27	2004.6- 2020.12	[34]
Lamont(OC)	36.6	-97.49	2008.7- 2020.12	[35]
Edwards(DF)	34.96	-117.88	2013.7- 2020.12	[36]
Caltech (CI)	34.14	-118.13	2012.9- 2020.12	[37]
East Trout Lake(ET)	54.36	-104.99	2016.10- 2020.9	[38]
Paris(PR)	48.85	2.36	2014.9- 2020.7	[39]
Orléans(OR)	47.97	2.11	2009.8- 2021.7	[40]
Bremen(BR)	53.1	8.85	2010.1- 2021.2	[41]
Garmisch(GM)	47.48	11.06	2007.7- 2021.4	[42]
Karlsruhe(KA)	49.1	8.44	2010.4- 2021.12	[43]
Bialystok(BI)	53.23	23.02	2009.3- 2018.10	[44]
Tsukuba(TK)	36.05	140.12	2011.8- 2019.9	[45]
Rikubetsu(RJ)	43.46	143.77	2013.11- 2019.9	[46]
Hefei(HF)	31.9	119.17	2015.9- 2018.10	[47]
Burgos(BU)	18.53	120.65	2017.3- 2020.3	[48]
Saga(JS)	33.24	130.29	2011.7- 2020.12	[49]

TABLE III
XGBOOST REGIONAL MODEL TEST ACCURACY

Region	r^2	MSE	RMSE	Soundings
Europe	0.94	0.83	0.91	402k
N.America	0.97	0.45	0.67	940k
Asia	0.98	0.03	0.17	130k

When it comes to optimizing the cost function, XGBoost adds binomial Taylor expansion and second-order derivative, contributing to anti-overfitting properties. In our experiments, the input features are retrieval state parameters, with the target values being TCCON observations. The training set accounts for 70%, while the test set comprises 30%. Through experiments, the model achieves sufficient convergence after 10 000 iterations. The test accuracy of each region is shown in Table III. The prediction value exhibits a strong correlation with the target value, with an accuracy is about 1 ppm and low uncertainty introduced by the model. Each OCO-2 measurement is input into the regional XGBoost model, and the predicted value represents the correction value sample, denoted as XCO_{2XGB}.

B. BC Considering Spatial Characteristics

At regional scales, there are discernible variations in the systematic errors between satellite measurements and each TCCON, reflecting the spatial characteristics of the bias. Satellite measurements that are co-located can be locally corrected by corresponding TCCON, while those without TCCON coverage cannot undergo such correction.

Considering the spatial correlation of the environmental measurement parameters, we hypothesized that the closer spatial proximity between satellite measurements and TCCON indicates a greater similarity in system bias characteristics.

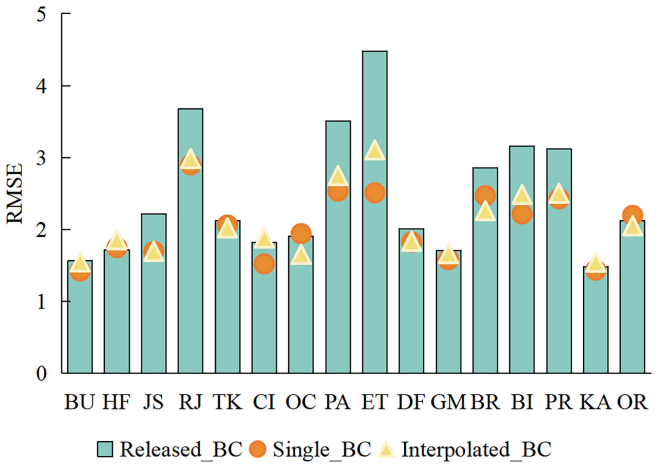


Fig. 3. Single_BC, Interpolated_BC are corrected based on each TCCON and the interpolation coefficients of other TCCON, respectively; Released_BC represents the released L2 BC product.

To demonstrate this, we compared the correction formula calculated at each TCCON with those interpolated from the other sites based on their spatial proximity. These two formulas were applied to each sounding, resulting in the correction known as "Single_BC" and "Interpolated_BC." The results show that their correction effects are highly similar, with an average difference of 0.17 ppm, and the bias is lower than the L2 BC product (Fig. 3). These suggest that using spatial correlation of systematic errors to correct satellite observations is feasible. By adopting this approach, we can correct each sounding using interpolated correction formulas from all TCCON sites, with the correction value being regarded as XCO_{2FI}.

C. BC Considering Temporal Characteristics

Long-term measurements conducted at TCCON sites have proven valuable in capturing important patterns of seasonal changes and annual growth in atmospheric CO_2 concentration [55]. The trigonometric polynomials and their related variants have been widely used in research to describe the temporal changes of atmospheric XCO_2 [56], [57], [58]. Ideally, the trends measured by satellite and ground-based are similar to overextended periods. Hence, the fitting formula of TCCON can be used to reconstruct the trends of satellite measurement collocated with TCCON. This study uses the Fourier series fitting method of trigonometric functions and divides the long-term TCCON measurement trend into interannual growth and seasonal cyclical changes

$$XCO_{2tccon,i} = A_i D + b_i Y. \tag{6}$$

Equation (6) is Fourier series fitting of long-term TCCON measurement sequence. Here, i represents a TCCON site, A_i refers to the Fourier coefficient, D is the day of year (DOY), b_i is the annual increase, and Y is the annual interval from the starting year

$$XCO_{2OCO_standard,i} = \frac{XCO_{2Raw,i} - FOOT[fp, mode] - FEATS[mode]_i}{TCCON_ADJUST[mode]_i}. \tag{7}$$

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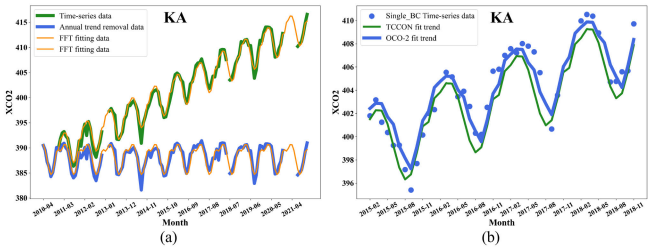


Fig. 4. Process of constructing the satellite reference benchmark. (a) Shows the trend fitting of the long-time series at the Karlsruhe site. (b) Shows the reconstruction of the continuous time series of satellite reference benchmarks.

Equation (7) denotes the BC for OCO-2 measurements co-located with TCCON, similar to (1), the correction coefficient is calculated independently by each site. Here, $XCO_{2Raw,i}$ and $XCO_{2OCO_standard,i}$ are the discrete OCO-2 raw retrieval value and correction value, respectively,

$$\text{adjust}_{\text{season},i} = \frac{XCO_{2OCO_standard,i,season}}{XCO_{2tccon,i,season}}. \tag{8}$$

The study designed a constant term $\text{adjust}_{\text{season},i}$ to adjust the differences in multisource observation which is a function of the season in (8)

$$XCO_{2oco,i} = (A_i D + b_i Y) * \text{adjust}_{\text{season},i}. \tag{9}$$

By substituting (6) and (8) into (9), $XCO_{2oco,i}$ represents the reconstructed continuous-time satellite reference benchmark. Multiple regional reference benchmarks are spatially interpolated at the satellite measurement location to obtain the correction value sample, denoted as XCO_{2RB}.

The fitting bias affects the reliability of the reference benchmark. Fig. 4 illustrates that Fourier trigonometric function fitting performs well in filling in continuous missing data. We analyzed that the fitting uncertainty of the satellite reference benchmark is 1.19 ppm, and the average RMSE is 1.16 ppm. Additionally, the $\text{adjust}_{\text{season},i}$ is larger in summer, indicating a higher uncertainty in satellite-to-ground measurement.

D. Bayesian Optimal Estimation

The RMSE of XCO_{2XGB}, XCO_{2RB}, and XCO_{2FI}compared to TCCON are 0.49, 1.00, and 0.86 ppm. These three methods are independent of each other and complementary. XCO_{2XGB} exhibits the highest accuracy but is limited to learning from TCCON-covered observations. In regions without TCCON coverage, it relies on the spatiotemporal correlations provided by the other two methods for supplementation. XCO_{2RB} is not adept at capturing occasional mutations, while XCO_{2XGB} is data-driven, and XCO_{2FI} is influenced by environmental observation parameters, thus serving as an effective supplement. XCO_{2FI} considers the spatial correlation of systematic errors to fill the correction for the TCCON-uncovered area, but its temporal generalization capability requires supplementation from the other two methods. BC based on Bayesian estimation can integrate the strengths of the three methods by combining the principle of systematic error correction with prior observations and handles uncertainty through the derivation of the posterior probability distribution of the target variable using the following equation:

$$p(\theta|X) = \frac{\pi(\theta) * p(X|\theta)}{\int_{-\infty}^{+\infty} \pi(\theta) p(X|\theta) d\theta}. \tag{10}$$

Applied to this research, $X = (XCO_{2XGB}, XCO_{2RB}, XCO_{2FI})$ is the sample of the correction values of XCO_2 drawn from the normal distribution $p(X\,|\,\theta) \sim N(\mu, \sigma_0^2)$, where μ is the optimal correction value to be sought. The prior distribution of μ is known to be $N(\gamma, \tau^2)$ from flux models. By substituting X and the prior distribution into (10), the posterior probability of μ can be obtained

$$\mu|X \sim N\left(\frac{n\bar{x}\sigma_0^{-2} + \gamma\tau^{-2}}{n\sigma_0^{-2} + \tau^{-2}}, \frac{1}{n\sigma_0^{-2} + \tau^{-2}}\right) \tag{11}$$

$$p\left(XCO_{2post}\right) = p^{(m)}(\hat{\mu}). \tag{12}$$

Estimating the optimal expectation using (11) provides the posterior correction value and uncertainty. Equation (12) is an iterator, that is, the posterior value from the previous loop is substituted as the prior value for the subsequent calculation. This iterative process is repeated m times ($m = 10$) to refine the posterior values.

TABLE IV

[1](A) ST-BC VERSUS TCCON. (B) ST-BC VERSUS MODELS. (C) ST-BC VERSUS ALL PROXIES

(A)

Region	RMSE/ppm	μ/ppm	σ/ppm
Europe	**1.28** (1.36)	**0.39** (0.62)	1.23 (**1.21**)
Asia	**1.20** (1.42)	**0.10** (0.64)	**1.19** (1.27)
N.America	**0.93** (1.12)	**0.10** (0.60)	0.95 (**0.94**)
ALL	**0.91** (0.98)	0.57 (**0.48**)	**0.71** (0.85)

(B)

Region	RMSE/ppm	μ/ppm	σ/ppm
Europe	**0.81** (1.37)	-0.08 (**-0.07**)	**0.81** (1.37)
Asia	**1.04** (1.39)	-0.40 (**-0.10**)	**0.95** (1.36)
N.America	**0.89** (1.18)	**-0.17** (-0.21)	**0.88** (1.16)
ALL	**0.86** (1.25)	-0.20 (**-0.17**)	**0.84** (1.24)

(C)

Product	RMSE/ppm	μ/ppm	σ/ppm
V10	1.07	-0.13	1.06
XGBoost [2]	1.02	-0.04	1.02
ST-BC [2]	0.88	-0.05	0.88

IV. RESULTS

A. Accuracy Evaluation

Using the ST-BC method, we produced regional bias-corrected daily XCO_2 products spanning from 2015 to 2018. To assess their accuracy, we compared these corrected products

[1]The performance of V10 under the same conditions is in brackets.
[2]The accuracy of V10 and XGBoost is cited by Keely et al. [32]

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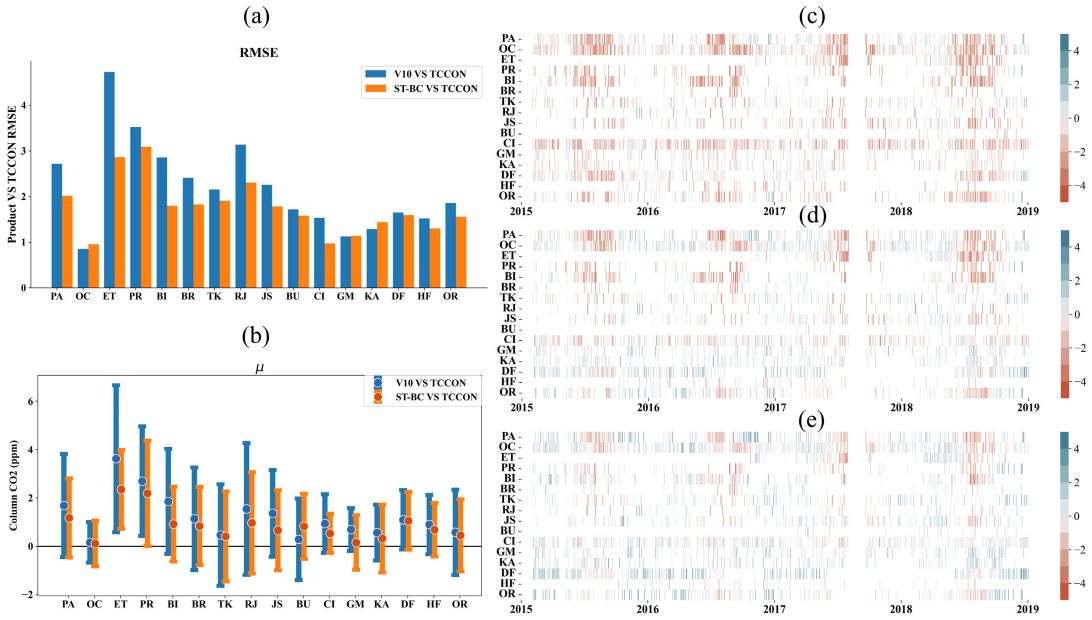


Fig. 5. Product versus TCCON. (a) Is the RMSE of the satellite-corrected product and TCCON. (b) Is the expectation (absolute value) and variance of the bias between the corrected product and TCCON. (c)–(e) Are the residuals of the raw retrieval, $L2$ correction, the ST-BC correction, and TCCON in the time series.

with measurements from TCCON sites and flux models in Table IV.

The ST-BC correction product demonstrates an overall land measurement accuracy of 0.88 ppm, with an uncertainty of 0.88 ppm. This accuracy is 0.19 ppm (18%) better than the V10 product and 0.14 ppm (14%) better than using only the XGBoost machine learning. The consistency of the correction products with ground-based observations and flux models has been improved. Regionally, the most notable improvement is observed in Asia, where the accuracy improves from 1.38 to 1.04 ppm. Europe and North America exhibit higher correction accuracies at 0.81 and 0.90 ppm, respectively. Specific sites such as PA and ET in North America, the BI in Europe, and the RJ in East Asia demonstrate more pronounced accuracy enhancements [Fig. 5(a) and (b)]. The residuals between the raw retrieval, V10 correction, and ST-BC correction products and TCCON gradually decrease [Fig. 5(c)–(e)]. This shows that the proposed method effectively reduces the regional bias, enhancing the accuracy of regional applications while reducing associated uncertainties

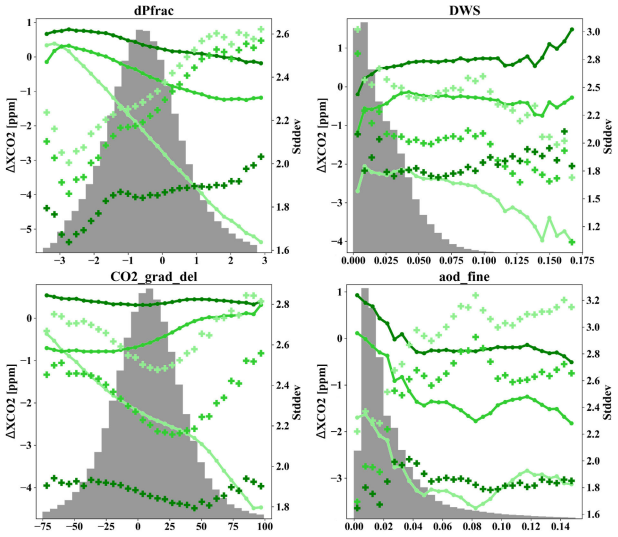


Fig. 6. Correlation between parameters and residuals. Light-green, lime–green, and green lines represent the raw retrieval, V10 correction, and ST-BC correction, respectively; dots represent variances.

B. Corrected Systematic Error

Evaluating the correlation between parameters and bias provides evidence to verify whether systematic errors have been eliminated. As shown in Fig. 6, the ST-BC correction method significantly weakens the correlation between the retrieval bias and the state parameters, outperforming the V10 method. This effect is more obvious for the dP_{frac} and DWS, which contribute the most to the bias. BC and quality filtering are complementary processes. A simple BC method implies strict quality filtering standards. However, the ST-BC method effectively compensates for the limitations of linear correction and performs well in removing the correlation of system parameters. This means that relatively loose filtering conditions can be applied to observation data that should not be eliminated.

The posterior variance of the Bayesian reflects the uncertainty of the estimate (Fig. 7). The uncertainty remains below 1 ppm for the majority of regions. Specifically, the Americas region exhibits relatively low uncertainty. However, in North China and certain observation orbits in Europe, the uncertainty ranges between 1 and 2 ppm. The impact of uncertainty on the results is discussed in Section V.

C. Compared With V10

Satellite products often exhibit underestimation at the regional scale, as evident from the predominantly negative

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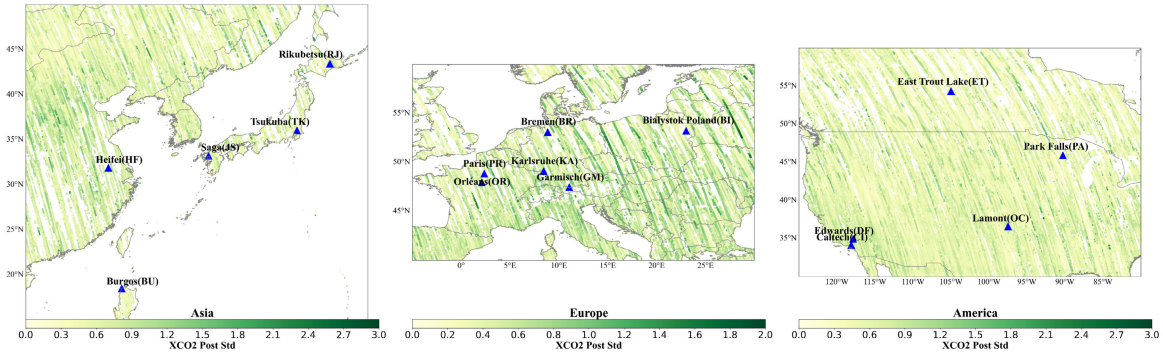
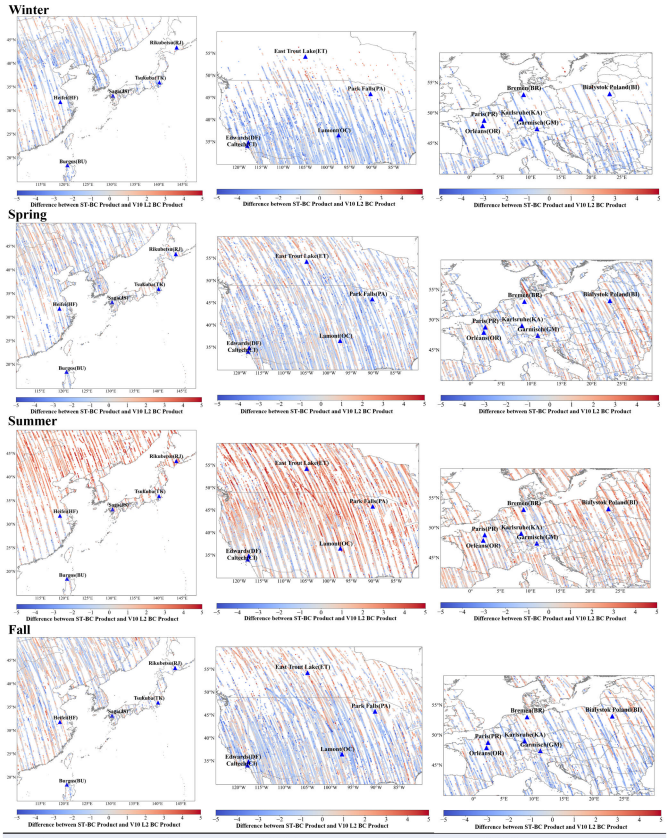


Fig. 7. Bayesian posterior uncertainty.



Region	DJF	MAM	JJA	SON	ALL
Europe	-0.89	-0.13	0.75	-0.48	0.02
Asia	-0.31	-0.02	1.71	0.12	0.31
N.America	-0.83	-0.32	0.93	-0.31	-0.04

Fig. 8. Differences between ST-BC product and V10.

residuals between V10 and TCCON measurements, as well as the scale adjustment factor typically being less than 1. This underestimation pattern follows similar seasonal patterns across the three regions (Fig. 8). During summer, a significant underestimation is observed, which constitutes the primary reason for the underestimation of satellite products. The underestimation of V10 products in the study areas of Asia, Europe, and North America is 1.71, 0.75, and 0.93 ppm on average. The underestimation is particularly prominent in northeastern China of Asia, southern Canada and east of the Rocky Mountains, the northern United States in North America, as well as the western East European Plain. In these areas, the underestimation is often around 3 ppm and continues into autumn. During spring and autumn, weaker overestimations are observed, averaging 0.30 ppm in Europe and North America, and a lower value of 0.05 ppm in Asia. In winter, the overestimation is relatively strong, with an average of 0.86 ppm in Europe and North America and a low value of 0.31 ppm in Asia. Despite V10 displaying regular overestimation and underestimation patterns across seasons, it is still close to the statistical distribution expectation of the correction product, indicating that the proposed method has mainly adjusted the uncertainty of regional bias.

V. Discussion

A. Seasonal Characteristics of the Bias

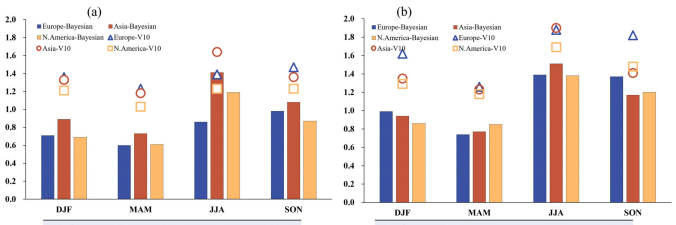
Summer climate changes often lead to active pressure belts or monsoon circulation, which activates the land ecosystem, and increases the frequency of CO_2 exchange between the ground and atmosphere. The relatively unstable atmospheric conditions introduce errors related to the signal and air mass during summer, resulting in higher bias [18], [59].

As depicted in Fig. 9(a), the correction results are more consistent with TCCON, particularly in summer, where the improvement effect is most prominent, reducing the bias by 0.78 ppm. The ST-BC method undoubtedly addresses the issue of large retrieval bias during summer to a certain extent. Fig. 9(b) shows that the overall accuracy of the seasons still follows the rule, which is low in summer and autumn and high in spring and winter. The highest and lowest accuracies are in spring (0.82 ppm) and summer (1.40 ppm). We also noticed some unsatisfactory results. For example, when comparing the spring correction results with TCCON, although the bias in Asia is reduced by 0.63 ppm, the accuracy in Europe and North America is slightly lower than V10 by about 0.03 and 0.08 ppm. Considering the overall accuracy, these fluctuations are deemed acceptable.

B. Bias Under Observation Modes

We also analyzed the performance of the correction results on different observation modes. In the Target mode, the correction results are more consistent with TCCON, with notable

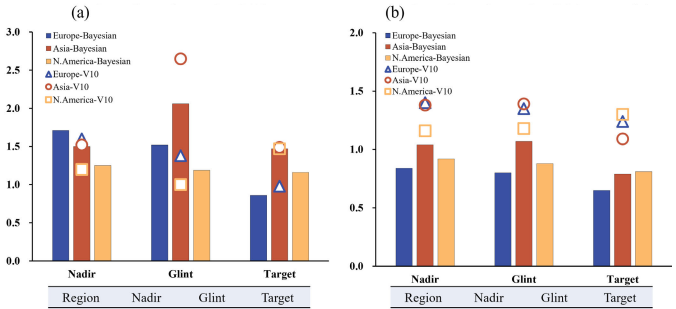
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Region	DJF	MAM	JJA	SON
Europe	**1.42** (1.70)	0.83 (**0.80**)	**1.95** (2.35)	**2.30** (2.36)
Asia	**1.55** (1.92)	**0.88** (1.51)	**2.42** (3.68)	**2.83** (2.85)
N.America	**0.97** (1.29)	1.28 (**1.20**)	**1.52** (2.23)	1.80 (**1.79**)
ALL	**1.10** (1.28)	**1.04** (1.19)	**1.55** (2.33)	**1.58** (1.92)

Region	DJF	MAM	JJA	SON
Europe	**0.99** (1.62)	**0.74** (1.26)	**1.39** (1.88)	**1.37** (1.82)
Asia	**0.94** (1.35)	**0.77** (1.23)	**1.51** (1.90)	**1.17** (1.41)
N.America	**0.86** (1.29)	**0.85** (1.18)	**1.38** (1.69)	**1.20** (1.48)
ALL	**0.90** (1.35)	**0.82** (1.20)	**1.40** (1.77)	**1.23** (1.53)

Fig. 9. Seasonal characteristics of the bias. (a) ST-BC product versus TCCON. (b) ST-BC product versus TCCON + Models.



Region	Nadir	Glint	Target
Europe	1.71 (**1.60**)	1.52 (**1.38**)	**0.86** (0.98)
Asia	**1.50** (1.52)	**2.06** (2.65)	**1.47** (1.49)
N.America	1.25 (**1.20**)	1.19 (**1.00**)	**1.16** (1.47)
ALL	**0.97** (1.02)	1.16 (**1.03**)	**1.16** (1.31)

Region	Nadir	Glint	Target
Europe	**0.84** (1.40)	**0.80** (1.35)	**0.65** (1.24)
Asia	**1.04** (1.38)	**1.07** (1.39)	**0.79** (1.09)
N.America	**0.92** (1.16)	**0.88** (1.18)	**0.81** (1.30)
ALL	**0.93** (1.26)	**0.90** (1.25)	**0.78** (1.26)

Fig. 10. Bias under different observation modes. (a) ST-BC product versus TCCON. (b) ST-BC product versus TCCON + Models.

improvements in North America (0.31 ppm), and higher accuracy in Europe (0.86 ppm) [Fig. 10(a)]. By considering the posterior uncertainty of the model, we identified that the high uncertainty in the areas represented by the DF, CI, and GM sites is the primary reason for the poorer performance of the Nadir and Glint patterns over Europe and North America. This uncertainty arises from differences between the prior data and TCCON or differences among the three correction samples. Although we evaluated in Section II-D that the bias between TCCON measurement and the flux model is minimal and filtered out prior data with large bias. However, the limited number of TCCON means that sites such as CI, which exhibit obvious differences with the flux model, cannot be easily filtered out. In terms of evaluating which measurement method, TCCON or the flux model, is closer to the truth, we prefer to consider ground measurements, such as TCCON, and treat the flux models as a proxy of the remaining space. The addition of flux models in Fig. 10(b) allows the accuracy to be more widely and reasonably validated. This further supports the superior performance of our method across all observation modes. Specifically, the Target mode achieves an accuracy of 0.78 ppm, while the accuracy in the Nadir and Glint modes is about 0.90 ppm.

C. Contribution of TCCON Sites

In this section, we evaluated the contribution of each site, that is, the impact of the sites on the accuracy of the results in the ST-BC method, and also reflected the difference in bias

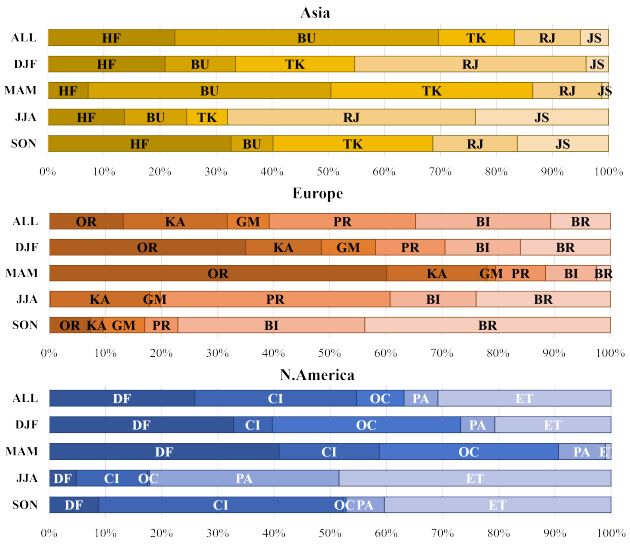


Fig. 11. Site contribution evaluation.

characteristics with other sites. In Section III-B, we defined the "Leave-One-Out" method to characterize site contribution, which means randomly deleting one site each time while retaining the remaining sites for modeling. The difference between the accuracy of all sites for modeling and the remaining sites for modeling as a percentage of the former represents the contribution of this site. The smaller percentage indicates a lower contribution, and vice versa. The results are shown in Fig. 11. The contribution of each site is often not fixed in seasons, which may be attributed to variations in environmental parameters and missing measurements. There is a general rule that the sites with larger differences in state parameters and greater differences in bias characteristics compared to other sites in the region tend to have higher contributions, such as BU in Asia, BI in Europe, and ET in North America. We hope that the contribution of regional sites can be valuable for guiding related ground observation applications.

D. Temporal Applicability

We extended the method to temporal analysis. We trained the modeling of the previous time (2015 to 2017) and applied it to the later time (2018) to verify the temporal applicability, which provided a certain degree of independence for the evaluation. Table V shows the regional accuracy of the ST-BC results in the training year and the prediction year. We found that the prediction accuracy and uncertainty are better than those of V10, indicating that the method has a certain temporal applicability. This suggests that the method can be utilized to update correction products when TCCON measurements are delayed. The long-term variations of XCO_2 are currently relatively stable. Considering potential future declines or abrupt changes in XCO_2, our model has an advantage in data-driven aspects compared to traditional single-formula approaches. However, compared with the sufficiently high model accuracy (0.75 ppm), the improvement of 2018 in prediction accuracy was not substantial. Upon analyzing the errors associated

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TABLE V
TIME APPLICABILITY

Region	Training Year (2015-2017)			Predicted Year (2018)		
	RMSE/ppm	μ/ppm	σ/ppm	RMSE/ppm	μ/ppm	σ/ppm
Europe	**1.41** (1.48)	0.69 (**0.65**)	**1.23** (1.33)	**0.74** (1.38)	**0.18** (-0.21)	**0.72** (1.36)
Asia	**1.31** (1.95)	**0.56** (0.87)	**1.18** (1.74)	**1.03** (1.40)	**-0.14** (-0.27)	**1.02** (1.37)
N.America	**0.74** (0.93)	**0.26** (0.36)	**0.69** (0.86)	**0.93** (1.24)	**0.17** (-0.38)	**0.91** (1.18)
ALL	**0.75** (0.90)	**0.25** (0.47)	**0.71** (0.77)	**0.92** (1.30)	**0.11** (-0.33)	**0.91** (1.26)

TABLE VI
PERFORMANCE OF THE MODEL AT NONMODELED SITES

Site	Lat/°	Lon/°	R^2	RMSE	μ	σ	sounding
SO	67.37	26.63	0.93 (0.88)	2.00 (2.85)	0.52 (1.39)	1.93(2.49)	46K
IZ	28.30	-16.48	0.93 (0.86)	1.25 (1.80)	-0.65 (-0.89)	1.07(1.57)	22K

with the three correction values, we found that the limitation of time-dependent BC is more pronounced due to the short duration of the training set.

VI. CONCLUSION AND FUTURE WORK

Improving the accuracy of satellite products is indeed a widely studied topic. Different from the research that mostly focuses on the fusion or assimilation of multisource data, our research focuses on the systematic errors in the product retrieval process itself. This involved minimizing errors induced by the limitations of the linear correction model. To achieve this, we integrate the systematic error correction theory, time series information from ground-based measurement, and regional bias characteristics, and establish a new regional correction method considering ST-BCs. The results demonstrate that the ST-BC method effectively reduces regional bias. The correction results obtained using the GGG2014 exhibit a bias of 0.83 ppm with the GGG2020, which is close to the accuracy of 0.80 ppm estimated by Taylor et al. [27]. We anticipate that the correction accuracy will be higher if the more accurate GGG2020 is used for modeling.

Although the study area covers most TCCON sites except for polar regions, achieving higher precision for application demands in regions with more diverse XCO_2 variability and greater human activity attention, we have also explored the method in regions lacking TCCON coverage, which will be a focus of our future work. In our previous research, we observed similar seasonal patterns and latitude dependencies in the spatial distribution of XCO_2 biases from satellite observations in the Northern Hemisphere, particularly in mid-latitude regions. Therefore, we attempt to extrapolate the bias patterns from three mid-latitude regions in this study to the broader Northern Hemisphere. We built the spatiotemporal distribution relationship between the ST-Bay corrected values and retrievals using deep learning. To simulate the model in regions without TCCON coverage, we selected the discrete IZ and SO sites from GGG2014 for validation, which were not involved in the modeling process. The experimental results are provided in Table VI. Clearly, whether at the low-latitude IZ or the high-latitude SO, our model's extrapolation to the Northern Hemisphere remains closer to ground truth compared to current methods. This attempt demonstrates the potential of this method, and we will continue to further research the model in subsequent research.

There are inevitably some inadequacies in some aspects. First, the fitting formula in Section III-C does not account for sudden changes in XCO_2 caused by natural activities with large-scale gradients. We also found that in the years when large El Niño occurred (such as 2015–2016), the fitting deviations in the lower latitude regions will fluctuate less. Additionally, the uncertainty introduced by the spatial interpolation method in Section III-B is another aspect, which may mean that the uncertainty of the prior value increases. In the current research, we focused on quality-filtered observations. So, improving the quality of "poor" measurements to increase data throughput is an important direction for future research. Last, the limitation of the number and distribution of regional TCCON sites restricts the ability to perform corrections on a global scale at present. As the number and coverage of TCCON sites increase, there is potential for the method to be applied in regional BC for ground sites in the future.

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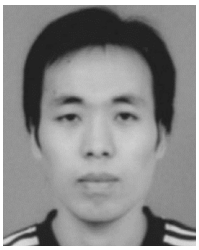
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